

A Project Report on

“Design & Manufacturing of Adaptive Cruise Control for Two Wheeler”

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CERTIFICATE

This is to certify that ***Mr. Mahesh Zirpe, Suyash Kadu, Atharva Thopate, Ganesh Sogam*** have successfully completed Project Stage – II entitled “**To Design & Manufacture of Adaptive Cruise Control for Two Wheeler**” under my supervision, in the partial fulfillment of Bachelor of Engineering - Mechanical Engineering of University of Pune.

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NOMENCLATURE

Symbols	Description	Unit
N	Newton	N
Lb	Pound – force	Lb
mm	Millimeter	mm
mpa	Mega pascal	mpa

ABBREVIATIONS

Symbols	Description
ACC	ADAPTIVE CRUISE CONTROL
PLA	POLYLACTIC ACID
UNO	1 in Italian
ACC	Adaptive cruise control
ECU	Electronic Control Unit
millisec	Milliseconds

ABSTRACT

This project is dedicated to designing and manufacturing an Adaptive Cruise Control (ACC) system tailored for two-wheeled vehicles. The primary goal is to enhance rider safety and convenience by employing a sophisticated control mechanism that integrates sensors, actuators, and intelligent algorithms. This allows for automatic speed adjustments based on the dynamic traffic environment.

The methodology involves a meticulous study of existing ACC systems, emphasizing the unique challenges and opportunities posed by the dynamics of two-wheeled vehicles in diverse traffic conditions. Selecting and integrating appropriate sensors, designing a compact actuator system, and formulating an efficient control algorithm are integral components of the project's approach.

Moving into the design and development phase, the project focuses on the seamless integration of sensors, the implementation of an effective actuator system, and the creation of a robust control unit. Addressing the intricacies of hardware and software integration ensures compatibility and reliability in real-world scenarios.

Testing and validation procedures encompass rigorous simulations, bench testing, and on-road trials. Performance metrics are established to evaluate the system's responsiveness, accuracy, and overall effectiveness. Results and discussions provide a comparative analysis with existing systems, highlighting the novel contributions of the designed ACC system to the two-wheeler domain.

In conclusion, the project not only presents a functional ACC system for two-wheelers but also discusses the encountered challenges, solutions implemented, and areas for future improvement. The recommendations section provides insights into enhancing the system's capabilities and potential applications beyond the scope of this project. The successful realization of this adaptive cruise control system for two-wheelers signifies a significant step forward in integrating cutting-edge technology to improve the safety and comfort of riders in urban traffic scenarios.